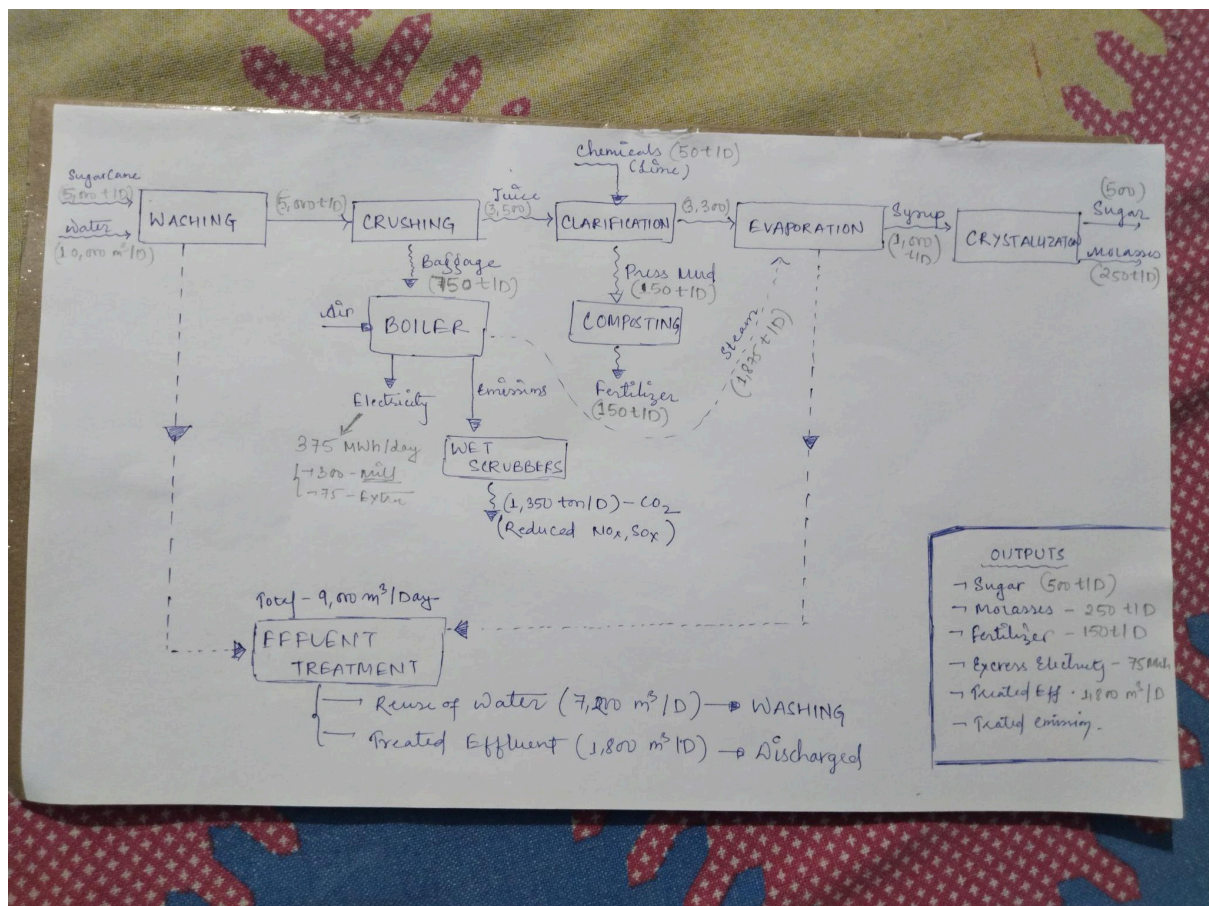




3. Material and Energy Balances

Assumptions: Based on the report of CPCB added as reference

- Daily sugarcane input: 5,000 tons
- Composition: 70% juice, 15% bagasse, 10% water, 5% impurities.
- Juice processing yields 10% sugar, 5% molasses, 3% Press mud.
- Energy: Bagasse provides 2.5 tons steam/ton burned; 1 ton bagasse generates 0.5 MWh electricity.

Material Balance (per day):

- **Inputs:**
 - Sugarcane: 5,000 tons
 - Water (washing, processing): 10,000 m³
 - Chemicals (lime, sulfur): 50 tons
- **Outputs:**
 - Sugar: 500 tons
 - Bagasse: 750 tons
 - Press mud: 150 tons
 - Molasses: 250 tons
 - Effluent: 9,000 m³ (90% of water input)
 - Emissions: CO₂, SO_x, NO_x from boilers

Energy Balance:

- **Inputs:**
 - Bagasse combustion: 750 tons × 2.5 tons steam/ton = 1,875 tons steam
 - Electricity generated: 750 tons × 0.5 MWh/ton = 375 MWh
- **Outputs:**
 - Steam for processing: 1,500 tons
 - Electricity for mill: 300 MWh
 - Excess electricity: 75 MWh

4. Waste Streams

1. **Solid Waste:**
 - Bagasse: 750 tons/day, fibrous residue.
 - Press Mud: 150 tons/day, organic sludge from clarification.
2. **Liquid Waste:**
 - Effluent: 9,000 m³/day, high BOD (2,500–3,000 mg/L), COD (4,000–6,000 mg/L), containing organic matter, sugars, and chemicals.
3. **Gaseous Waste:**
 - Boiler Emissions: CO₂ (1,500 tons/day), SO_x, NO_x from bagasse combustion.

- Current Handling

- **Bagasse:** Burned in boilers for steam/power (80%), surplus landfilled or sold as low-value fuel.
- **Press Mud:** Disposed in landfills or used as low-grade fertilizer.
- **Effluent:** Treated via anaerobic digestion and aeration ponds, but often partially discharged into water bodies, causing pollution.
- **Emissions:** Controlled with electrostatic precipitators, but SOx/NOx emissions remain significant

5. Optimization

1. **Bagasse Valorization:**
 - Convert 50% surplus bagasse (150 tons/day) to bioethanol or biogas via anaerobic digestion.
 - Use briquetting for remaining bagasse to produce high-value fuel pellets.
2. **Press Mud Utilization:**
 - Compost press mud with microbial additives to produce high-quality organic fertilizer.
3. **Effluent Management:**
 - Implement advanced anaerobic digestion followed by membrane bioreactors to reduce BOD/COD by 95%, enabling water reuse.
4. **Emission Control:**
 - Installed wet scrubbers to reduce SOx/NOx emissions by 80%.
 - Optimize boiler efficiency to reduce CO₂ emissions by 10%.

- Before/After Comparison

Waste Stream	Baseline (tons/day or m ³ /day)	Optimized (tons/day or m ³ /day)	Reduction (%)
Bagasse (landfilled)	150	0 (valorized)	100%
Press Mud (landfilled)	150	0 (composted)	100%
Effluent (discharged)	9,000	1,800 (80% reused)	80%
CO ₂ Emissions	1,500	1,350	10%

Resource Efficiency: AS PER REPORT OF CPCB

- Water reuse: 7,200 m³/day saved.
- Excess electricity: Increases from 75 MWh to 100 MWh/day due to biogas production.

6. Life Cycle Assessment (LCA) and Product Carbon Footprint (PCF)

i) Goal and Scope

Goal: Identify environmental impacts of sugar production and evaluate optimization strategies.

Scope: Cradle-to-gate (sugarcane cultivation to sugar production), including waste management.

ii) Life Cycle Inventory (LCI)

- **Inputs** (per ton sugar):
 - Sugarcane: 10 tons
 - Water: 20 m³
 - Energy: 0.6 MWh (from bagasse)
 - Chemicals: 0.1 tons
- **Outputs:**
 - Sugar: 1 ton
 - Wastes: 1.5 tons bagasse, 0.3 tons press mud, 0.5 tons molasses, 18 m³ effluent
 - Emissions: 3 tons CO₂eq (cultivation, processing, boilers)

iii) Life Cycle Impact Assessment

Baseline Impacts:

- Global Warming Potential (GWP): 3,000 kg CO₂eq/ton sugar, driven by cultivation (50%), boilers (30%), and waste disposal (20%).
- Water Use: 20 m³/ton, high due to washing and processing.
- Eutrophication: High from effluent discharge (BOD/COD).
- Land Use: Significant due to sugarcane cultivation.

Optimized Impacts:

- GWP: Reduced to 2,400 kg CO₂eq/ton (20% reduction) due to biogas production, emission controls, and waste valorization.
- Water Use: Reduced to 4 m³/ton (80% reduction) via effluent reuse.

- Eutrophication: Near-eliminated with advanced effluent treatment.
- Land Use: Unchanged, as cultivation remains constant.

iv) Life Cycle Interpretation

- Biogas and waste valorization significantly reduce GWP and improve energy efficiency.
- Effluent treatment and reuse drastically cut water usage and eutrophication potential.
- Cultivation practices are the largest impact contributor, needing further improvement (e.g., low-emission fertilizers).
- Land use remains a fixed impact, highlighting the need for higher crop yield or alternate sourcing.
- Integrated optimization across waste, water, and emissions offers a sustainable sugar production pathway.

v) Product Carbon Footprint (PCF)

- **Baseline PCF**: 3,000 kg CO₂eq/ton sugar.
- **Optimized PCF**: 2,400 kg CO₂eq/ton sugar, aligning with industry benchmarks for sustainable sugar production.

7. Cost-Benefit Analysis (CBA)

i) Raw Material Costs

- Sugarcane:
 - Quantity: 5,000 tons/day.
 - Cost: \$30/ton (based on regional market prices in India, as referenced in Sachdeva et al., 2024).
 - Annual Cost (300 operating days): $5,000 \text{ tons/day} \times \$30/\text{ton} \times 300 \text{ days} = \textbf{\$45,000,000/year}$.
- Water:
 - Quantity: 10,000 m³/day (for washing and processing).
 - Cost: \$0.10/m³ (regional water pricing, per CPCB Charter).
 - Annual Cost: $10,000 \text{ m}^3/\text{day} \times \$0.10/\text{m}^3 \times 300 \text{ days} = \textbf{\$300,000/year}$.
- Chemicals (lime, sulfur, etc.):
 - Quantity: 50 tons/day.
 - Cost: \$200/ton
 - Annual Cost: $50 \text{ tons/day} \times \$200/\text{ton} \times 300 \text{ days} = \textbf{\$3,000,000/year}$.
- Total Raw Material Costs:
 - $\$45,000,000 \text{ (sugarcane)} + \$300,000 \text{ (water)} + \$3,000,000 \text{ (chemicals)} = \textbf{\$48,300,000/year}$.

ii) Costs after Optimization - (all data taken from the references mentioned)

- **Capital Costs** :

- Biogas Plant – \$2,000,000 (for converting 150 tons/day surplus bagasse to biogas).
- Membrane Bioreactors – \$1,500,000 (for effluent treatment, reducing BOD/COD by 95%).
- Wet Scrubbers — \$500,000 (for reducing SOx/NOx emissions by 80%).
- Composting Facility — \$300,000 (for converting 150 tons/day press mud to fertilizer).
- Total Capital Cost — $\$2,000,000 + \$1,500,000 + \$500,000 + \$300,000 = \$4,300,000$.
- **Operational Costs (annual):**
 - Maintenance, labor, and chemicals for biogas plant, bioreactors, scrubbers, and composting: \$500,000/year..

iii) Analysis

Benefits

- **Operational Savings (annual):**
 - Water Reuse — $7,200 \text{ m}^3/\text{day reused} \times \$0.10/\text{m}^3 \times 300 \text{ days} = \$216,000/\text{year}$.
 - Reduced Waste Disposal — $300 \text{ tons/day (bagasse + press mud)} \times \$3/\text{ton} \times 300 \text{ days} = \$270,000/\text{year}.$
 - Total Savings: $\$216,000 + \$270,000 = \$486,000/\text{year}$.
- **Revenue (annual):**
 - Biogas Electricity — $25 \text{ MWh/day excess (from biogas)} \times \$0.08/\text{kWh} \times 300 \text{ days} = \$600,000/\text{year}$.
 - Organic Fertilizer Sales — $150 \text{ tons/day press mud compost} \times \$10/\text{ton} \times 300 \text{ days} = \$450,000/\text{year}$.
 - Total Revenue: $\$600,000 + \$450,000 = \$1,050,000/\text{year}$.
- **Total Annual Benefits:** $\$486,000 \text{ (savings)} + \$1,050,000 \text{ (revenue)} = \$1,536,000/\text{year}$.

Economic Feasibility

- **Payback Period:** $\$4,300,000 \text{ (capital cost)} \div \$1,536,000/\text{year (benefits)} \approx \mathbf{2.8 \text{ years}}$.
- **Net Present Value (NPV):** Assuming a 10% discount rate over 10 years, NPV = **\$6,238,000** (calculated using discounted cash flows, indicating strong profitability).

8. Final Recommendations

1. **Process Changes:**
 - Implement biogas production from bagasse and advanced effluent treatment to reduce waste and emissions.
 - Optimize boiler efficiency and install wet scrubbers for emission control.
2. **Recycling/Valorization:**

- Convert press mud to organic fertilizer and surplus bagasse to biogas or bioethanol.
 - Reuse 80% of treated effluent in milling operations.
3. **Trade-offs:**
- **Environmental:** Significant reductions in GWP, water use, and eutrophication, but cultivation impacts remain high.
 - **Economic:** High initial costs offset by rapid payback and long-term savings/revenue.
4. **Implementation:**
- Phase 1: Install biogas and effluent treatment systems (Year 1).
 - Phase 2: Develop composting and emission control systems (Year 2).
 - Monitor LCA/PCF annually to track progress.

9. References

- Central Pollution Control Board — Charter for Water Recycling & Pollution Prevention in Sugar Industries. https://cpcb.nic.in/ngrba/charter_sugar.pdf
- Central Pollution Control Board — Charter 2.0 for Sugar Industries. <https://cpcb.nic.in/ngrba/Charter-2.0-Sugar.pdf>
- Sachdeva, A., et al. (2024). Waste-to-energy —The recycling and reuse of sugar industry waste for different value-added products such as bioenergy in selected countries – a critical review. <https://www.researchgate.net/publication/379543482>